a polypeptide chain is represented by a sequence of three nucleotides in the RNA known as the genetic code. RNA transfers genetic message from nucleus to the cytoplasm. DNA is always present in the nucleus and synthesis is also confined to the nucleus

Jacob and Monod proposed the classical model of *Lac* operon to explain gene expression and regulation in *E. coli*. In *lac* operon a polycistronic structural gene is regulated by a common promoter and regulator. It is an example of negative control of transcription initiation.

Human genome project, a mega project was aimed to sequence every gene in the human genome. Polymerase chain reaction is an *in vitro* method of synthesis of nucleic acids wherein, a specific DNA segment is amplified rapidly without concomitant replication of the rest of the DNA molecule. DNA fingerprinting is a technique to identify variations in individuals of a population at the DNA level. It has immense applications in the field of forensic analysis, pedigree analysis, anthropological studies, and conservation of wild life.

Evaluation



1. Hershey and Chase experiment with bacteriophage showed that
   a) Protein gets into the bacterial cells
   b) DNA is the genetic material
   c) DNA contains radioactive sulphur
   d) Viruses undergo transformation

2. DNA and RNA are similar with respect to
   a) Thymine as a nitrogen base
   b) A single-stranded helix shape
   c) Nucleotide containing sugars, nitrogen bases and phosphates
   d) The same sequence of nucleotides for the amino acid phenyl alanine

3. A mRNA molecule is produced by
   a) Replication **b) Transcription**
   c) Duplication d) Translation

4. The total number of nitrogenous bases in human genome is estimated to be about
   a) 3.5 million b) 35000
   c) 35 million **d) 3.1 billion**

5. *E. coli* cell grown on ^{15}N medium are transferred to ^{14}N medium and allowed to grow for two generations. DNA extracted from these cells is ultracentrifuged in a cesium chloride density gradient. What density distribution of DNA would you expect in this experiment?
   (a) One high and one low density band.
   (b) One intermediate density band.
   (c) One high and one intermediate density band.
   (d) One low and one intermediate density band.

6. What is the basis for the difference in the synthesis of the leading and lagging strand of DNA molecules?
   (a) Origin of replication occurs only at the 5' end of the molecules.
   (b) DNA ligase works only in the 3' → 5' direction.
   (c) DNA polymerase can join new nucleotides only to the 3' end of the growing stand.
   (d) Helicases and single-strand binding proteins that work at the 5' end.

7. Which of the following is the correct sequence of event with reference to the central dogma?
   (a) Transcription, Translation, Replication
   (b) Transcription, Replication, Translation
   (c) Duplication, Translation, Transcription
   (d) Replication, Transcription, Translation

TN_GOVT_XII_Zoology_chapter5.indd 87 24-02-2020 4.01.39 PM